

Subnetting: -

One goal of a subnet is to split a large network into a grouping of smaller, interconnected networks to help minimize traffic.

Advantages of Subnetting:

- **Reduce Network Traffic**
- **Optimized Network Performance**
- **Simplified Management**
- **Facilitated Spanning of Large geographical distance.**
- **Bandwidth Utilization will increase**

How to Prevent wastage of IP Address?

Because of IPV4 addresses are Limited we have some methods to save the IP Address: -

- **Using Private IP Addresses**
- **Using the Subnetting**
- **IPV6 Addresses**

Why Need Subnetting?

<https://www.youtube.com/@techwithravish>

Host Available in Class A, Class B and Class C:

Class C: 256 Fixed

Class B: 65,536 Fixed

Class A: 16,777,216 Fixed

Let us take some example,

Need 2 IP Address but provide 256 IP, Waste 254

Need 4 IP Address but provide 256 IP, Waste 250

Need 60 IP Address but provide 256 IP, Waste 296

From 0-254 Address 256 IP will be allotted

This is result of wastage of IP

Why Need Subnetting?

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In Case of Class B, Number of Host = 65536

so,

If Need 500 IP Address but provide 65536 IP

If Need 1000 IP Address but provide 65536 IP,

If Need 2000 IP Address but provide 65536 IP,

**Means if need any IP in between 255 – 65536 Addresses,
65536 IP Addresses will be allotted**

This is result of Lots of wastage of IP

How to Allot 60 IP Address:

Allotting 60 IP Address from 192.168.1.0 Network of Class C,

192.168.1.0

- Network IP

192.168.1.1

192.168.1.2

.

192.168.1.60

Requirement

192.168.1.61

.

192.168.1.254

Wastage of 194 IP

192.168.1.255

– Broadcast IP

Class C Subnetting

Email: Need 60 IP Addresses.

Ans: We will use the Class C network 192.168.1.0

Step 1: Find the Host bit “n”:

$(2^n) - 2 \geq 60$ Satisfy Equation by putting

$(2^6) - 2 \geq 60$ values of N as 0 1 2 3 4 5 6 7 8

$(64) - 2 \geq 60$

$62 \geq 60$, here $n = 6$

Last 6 Bits of 32 Bits will be used as the host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

- According to the Step 1, there are total 6 Host bits
- Host Bits Represented by 0

Default Subnet Mask: **255.255.255.0**

11111111.11111111.11111111.00000000

Moving Right to Left and turn 6 bits "0" and Remaining 1

11111111.11111111.11111111.11000000

Customized Subnet Mask: **255.255.255.192**

Count the 1s in the above bit format,

CIDR Value = /26

Step 3: Block Size = $2^n = 2^6 = 64$

Allotting the IP Address:

192.168.10.0 Network IP

192.168.10.1/26

192.168.10.2/26

192.168.10.3/26 Valid IP Addresses = 62

.

.

192.168.10.62/26

192.168.10.63 Broadcast IP

Wastage = $62 - 60 = 2$ (This wastage cannot be saved)

Example 2:

Email: Need 30 IP Addresses.

Ans: We will use the Class C network 192.168.1.0

Step 1: Find the Host bit “n”:

$(2^n) - 2 \geq 30$ Satisfy Equation by putting

$(2^5) - 2 \geq 30$ values of N as 0 1 2 3 4 5 6 7 8

$(32) - 2 \geq 30$

$30 \geq 30$, here $n = 5$

Last 5 Bits of 32 Bits will be used as the host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

- According to the Step 1, there are total 5 Host bits
- Host Bits Represented by 0

Default Subnet Mask: **255.255.255.0**

11111111.11111111.11111111.00000000

Moving Right to Left and turn 5 bits "0" and Remaining 1

11111111.11111111.11111111.11100000

Customized Subnet Mask: **255.255.255.224**

Count the 1s in the above bit format,

CIDR Value = /27

Step 3: Block Size = $2^n = 2^5 = 32$

Allotting the IP Address:

192.168.10.0 Network IP

192.168.10.1/27

192.168.10.2/27

192.168.10.3/27 Valid IP Addresses = 30

.

.

192.168.10.30/27

192.168.10.31 Broadcast IP

Wastage = $30 - 30 = 0$ (No Wastage)

Example 3:

Email: Need 2 IP Addresses.

Ans: We will use the Class C network 192.168.1.0

Step 1: Find the Host bit “n”:

$(2^n) - 2 \geq 2$ Satisfy Equation by putting

$(2^2) - 2 \geq 2$ values of N as 0 1 2 3 4 5 6 7 8

$(4) - 2 \geq 2$

$2 \geq 3$, here $n = 2$

Last 2 Bits of 32 Bits will be used as the host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

- According to the Step 1, there are total 2 Host bits
- Host Bits Represented by 0

Default Subnet Mask: **255.255.255.0**

11111111.11111111.11111111.00000000

Moving Right to Left and turn 2 bits “0” and Remaining 1

11111111.11111111.11111111.11111100

Customized Subnet Mask: **255.255.255.252**

Count the 1s in the above bit format,

CIDR Value = /30

Step 3: Block Size = $2^n = 2^2 = 4$

Allotting the IP Address:

192.168.10.0 Network IP

192.168.10.1/30

192.168.10.2/30 Valid IP Addresses = 2

192.168.10.3 Broadcast IP

Wastage = $2 - 2 = 0$ (No Wastage)

Class B Subnetting:

Email: Need 500 IP Addresses.

Ans: We will use the Class B network 172.17.0.0

Step 1: Find the Host bit “n”:

$(2^n) - 2 \geq 500$ Satisfy Equation by putting

$(2^9) - 2 \geq 500$ values of N as 0 1 2 3 4 5 6 7 8

$(512) - 2 \geq 500$

$510 \geq 500$, here $n = 9$

Last 9 Bits of 32 Bits will be used as the host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

- According to the Step 1, there are total 9 Host bits
- Host Bits Represented by 0

Default Subnet Mask: **255.255.0.0**

11111111.11111111.00000000.00000000

Moving Right to Left and turn 9 bits "0" and other as 1

11111111.11111111.11111110.00000000

Customized Subnet Mask: **255.255.254.0**

Count the 1s in the above bit format,

CIDR Value = /23

Step 3: Block Size = $2^n = 2^9 = 512$

Allotting the IP Address:

172.17.0.0/23 Network IP

172.17.0.1/23

.

172.17.0.255/23 Valid IP Addresses = 510

172.17.1.0/23

.

172.17.1.254/23

172.17.1.255/23 Broadcast IP

Wastage = $510 - 500 = 10$ (This wastage cannot be saved)